Kyle Charlton

kyle@kcharlton.dev | github.com/kylejcharlton

Skills

Backend: C++, CUDA, C#, Node.js, Python
Frontend: JS, React, WebGL
Infrastructure: Docker, Kubernetes, Jenkins, Rancher

Experience

Warranty Service Manager, SOLE Fitness – Salt Lake City, UT — August 2024 – Present

- Built an initial **workflow-driven service portal** using **Monday.com and Make.com**, implementing custom logic, automations, and state transitions to support contractor assignment, job acceptance, progress transparency, and order closeout.
- Collaborated closely with a software engineer to guide development of a **C# Blazor web application**, providing detailed specifications, rapid feedback cycles, and rigorous user acceptance testing to ensure correctness and usability.
- Designed and executed a **full data migration strategy** from the legacy workflow system to the new application, preparing and validating data using **Python scripts** to ensure continuity and data integrity.
- Delivered a production-ready system **from scratch in 4 months**, replacing third-party tooling and reducing recurring SaaS costs by **$40K annually** while improving reliability, performance, and extensibility.

DevOps Intern, OpenText – Provo, UT — August 2022 – December 2023

- Led a redesign of an internal **DevOps dashboard** used by multiple engineering teams to visualize Jenkins pipeline status and build artifacts, improving build visibility and reducing time spent diagnosing CI failures
- Designed and implemented **backend APIs** to aggregate Jenkins pipeline data, test results, and artifacts, enabling real-time updates and consistent data access across teams and tooling.
- Enhanced support for **test-driven development (TDD)** by surfacing granular test pass/fail metrics and trends directly in the dashboard, helping teams identify flaky tests and regressions earlier in the development cycle.
- Deployed and maintained the application on **Kubernetes via Rancher**, contributing to containerized deployments, configuration updates, and rollout monitoring in a multi-team environment.

Software Intern, nView medical – Salt Lake City, UT — April 2020 – October 2021

- Improved 3D X-ray reconstruction accuracy by enhancing a **C++/CUDA** reconstruction pipeline to model real-world X-ray panel readout behavior, reducing motion and timing artifacts in clinical scenarios.
- Developed a **real-time tool-tracking visualization** system using JavaScript and WebGL that integrated camera input and reflective bead tracking to accurately align the rendered camera view with the true physical position of surgical tools.
- Designed and implemented a **WebGL-based UI** with live camera tracking to visualize the conical region of interest (ROI) for upcoming X-ray captures, giving clinicians real-time feedback on what anatomy would be imaged next.

Projects

MyPetTag - A Smart Pet Tag — github.com/dummio/MyPetTag

- Built a **smart pet identification system** that links uniquely generated **QR codes and NFC tags** affixed to a pet's collar with a digital profile accessible via smartphone scan, enabling quick contact of owners if pets are found.
- Engineered automated **owner notification mechanisms** to alert pet owners when their pet's tag was scanned, including time and location context to accelerate recovery efforts.
- Designed and developed a **responsive web interface** using React for managing pet profiles, allowing owners to update animals' details, care instructions, and emergency info in a secure, user-friendly environment.

Education

University of Utah – BS in Software Development — December 2024